

Predictive Modeling of Epileptic Seizure Detection Using Machine Learning

Rahul Rajput

2026-04-16

1. Introduction

The dataset that is used in this study is BEED of UCI repository, which includes EEG signal attributes (X1 - X16) and a target variable (y) that is a measure of seizure activity. The objective is to make predictive models that are capable of correctly classifying seizure and non-seizure cases to assist in clinical decision-making.

2. Methods and Analysis

The data set was initially purged of any missing data to allow proper analysis and the target variable was turned into a simple binary data (seizure vs non-seizure) to facilitate classification. This was followed by dividing the data into 80 percent training and 20 percent testing, which will enable the model to learn efficiently and also to evaluate the performance on the unseen data fairly. This study employed three models: Logistic Regression as an interpretable and simple baseline, Random Forest to learn the intricate trends in the data, and XGBoost to have a better prediction performance due to the use of advanced boosting models. Logistic Regression was employed as a baseline because it is simple and easy to interpret. Random Forest was chosen because it assumes the presence of non-linear relationships, and the ability to handle features that are correlated with each other, without involving feature scaling. XGBoost was added to test the hypothesis of whether gradient boosting algorithms could further improve predictive accuracy by sequential learning.

2.1 Libraries

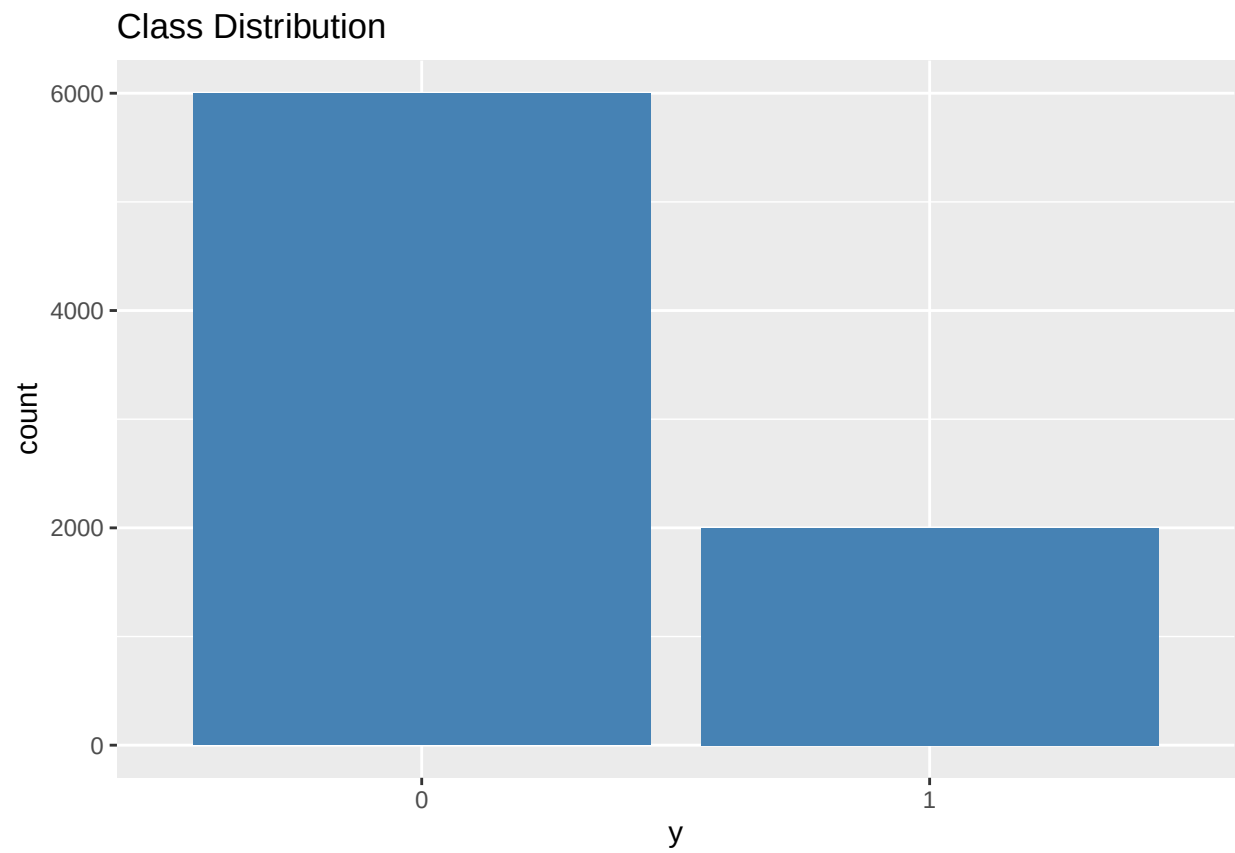
```
library(here)
library(tidyverse)
library(caret)
library(ranger)
library(xgboost)
library(pROC)
library(knitr)
```

2.2 Load Data

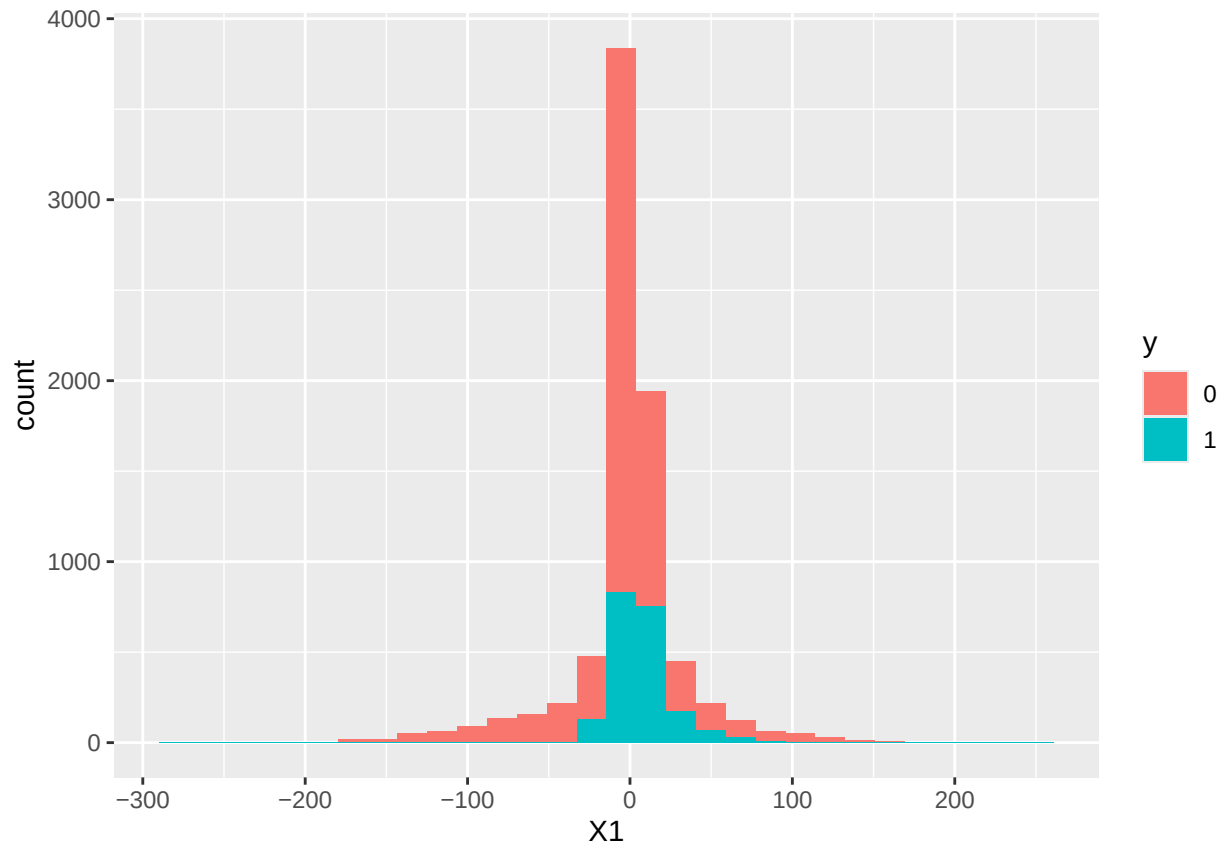
2.3 Preprocessing

2.4 Data Visualization

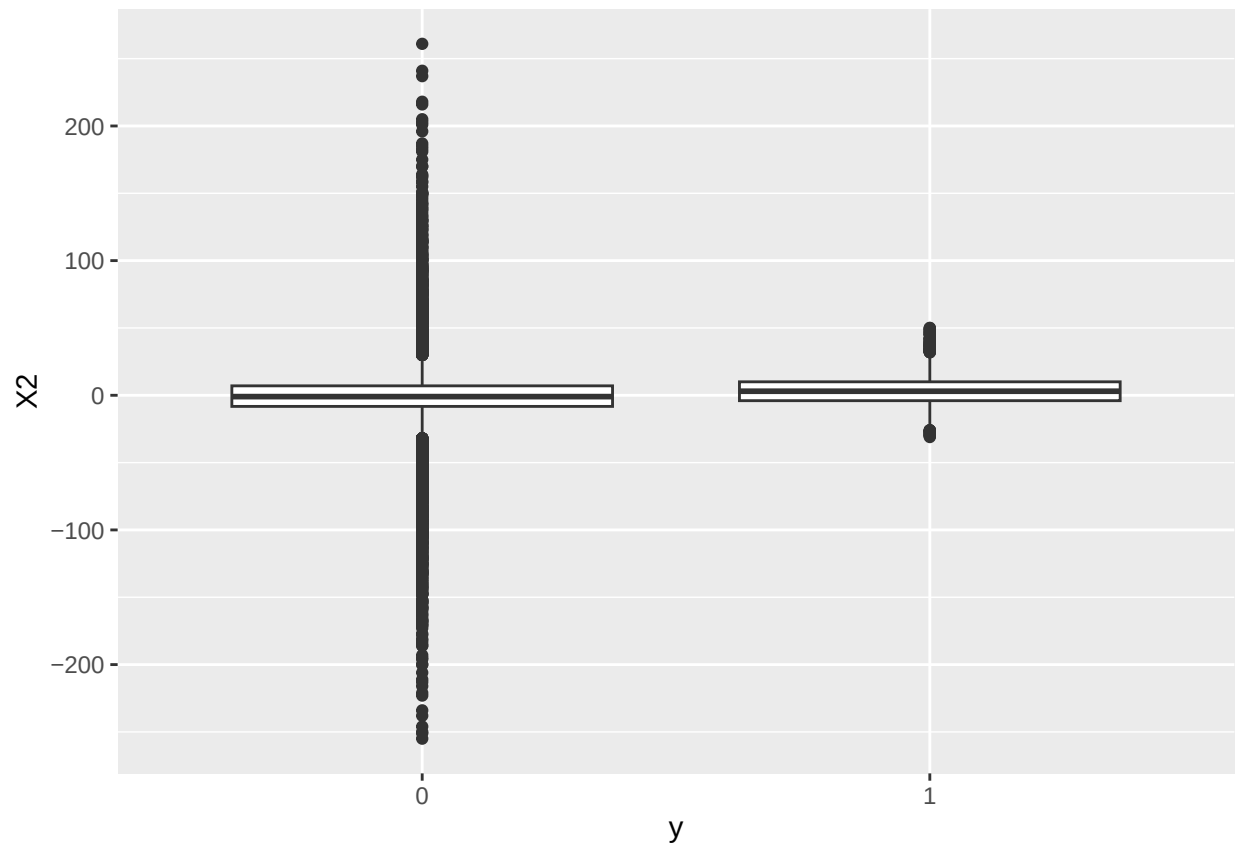
Class Distribution



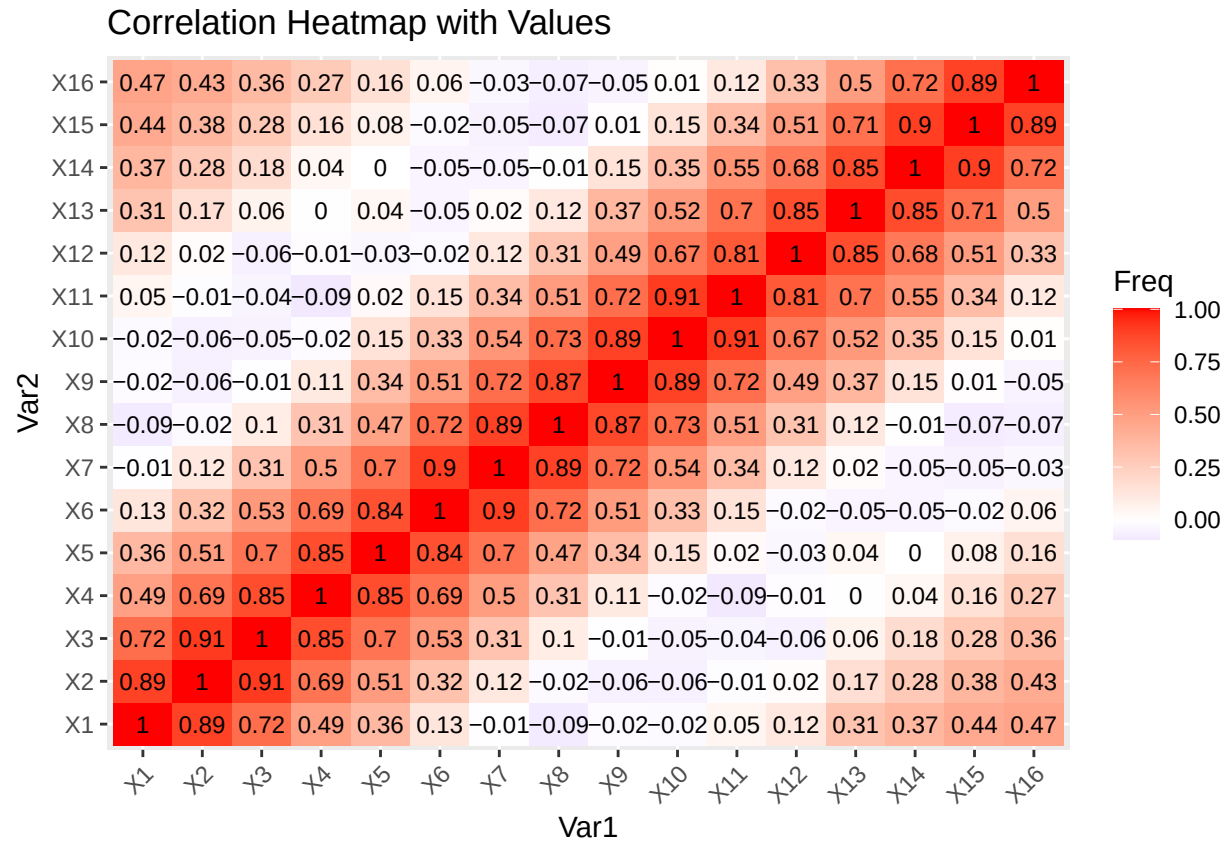
Histogram



Boxplot



Correlation Heatmap



Correlation heatmap reveals that there are moderate to high correlation coefficients between a number of EEG features, indicating the redundancy in signal characteristics. This justifies the use of tree-based models such as Random Forest which is resistant to multicollinearity and capable of effectively capturing complex interactions between features.

2.5 Train-Test Split

Stratified sampling to maintain the distribution of classes was used to divide the dataset into 80% training and 20% testing. This will ensure that the models are trained using adequate data and yet they are reliably evaluated using unknown samples.

2.6 Logistic Regression

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 1150  285
##           1   50  115
##
##           Accuracy : 0.7906
##           95% CI : (0.7699, 0.8103)
##           No Information Rate : 0.75
```

```

##      P-Value [Acc > NIR] : 7.48e-05
##
##              Kappa : 0.3057
##
## Mcnemar's Test P-Value : < 2.2e-16
##
##      Sensitivity : 0.9583
##      Specificity : 0.2875
##      Pos Pred Value : 0.8014
##      Neg Pred Value : 0.6970
##      Prevalence : 0.7500
##      Detection Rate : 0.7188
##      Detection Prevalence : 0.8969
##      Balanced Accuracy : 0.6229
##
##      'Positive' Class : 0
##

```

2.7 Random Forest

```

## Confusion Matrix and Statistics
##
##      Reference
## Prediction    0    1
##      0 1195    27
##      1     5   373
##
##      Accuracy : 0.98
##      95% CI : (0.9719, 0.9863)
##      No Information Rate : 0.75
##      P-Value [Acc > NIR] : < 2.2e-16
##
##      Kappa : 0.9457
##
## Mcnemar's Test P-Value : 0.0002054
##
##      Sensitivity : 0.9958
##      Specificity : 0.9325
##      Pos Pred Value : 0.9779
##      Neg Pred Value : 0.9868
##      Prevalence : 0.7500
##      Detection Rate : 0.7469
##      Detection Prevalence : 0.7638
##      Balanced Accuracy : 0.9642
##
##      'Positive' Class : 0
##

```

2.8 XGBoost

```

## Confusion Matrix and Statistics
##

```

```
##           Reference
## Prediction    0    1
##           0 1189   30
##           1   11  370
##
##           Accuracy : 0.9744
##           95% CI : (0.9654, 0.9815)
##           No Information Rate : 0.75
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.9306
##
## Mcnemar's Test P-Value : 0.004937
##
##           Sensitivity : 0.9908
##           Specificity : 0.9250
##           Pos Pred Value : 0.9754
##           Neg Pred Value : 0.9711
##           Prevalence : 0.7500
##           Detection Rate : 0.7431
##           Detection Prevalence : 0.7619
##           Balanced Accuracy : 0.9579
##
##           'Positive' Class : 0
##
```

2.9 Model Comparison

```
get_metrics <- function(cm){
  c(
    Accuracy = as.numeric(cm$overall["Accuracy"]),
    Precision = as.numeric(cm$byClass["Precision"]),
    Recall = as.numeric(cm$byClass["Recall"])
  )
}

results <- data.frame(
  Model = c("Logistic Regression", "Random Forest", "XGBoost"),
  rbind(get_metrics(cm_lr), get_metrics(cm_rf), get_metrics(cm_xgb))
)

knitr::kable(results, caption = "Model Comparison Table")
```

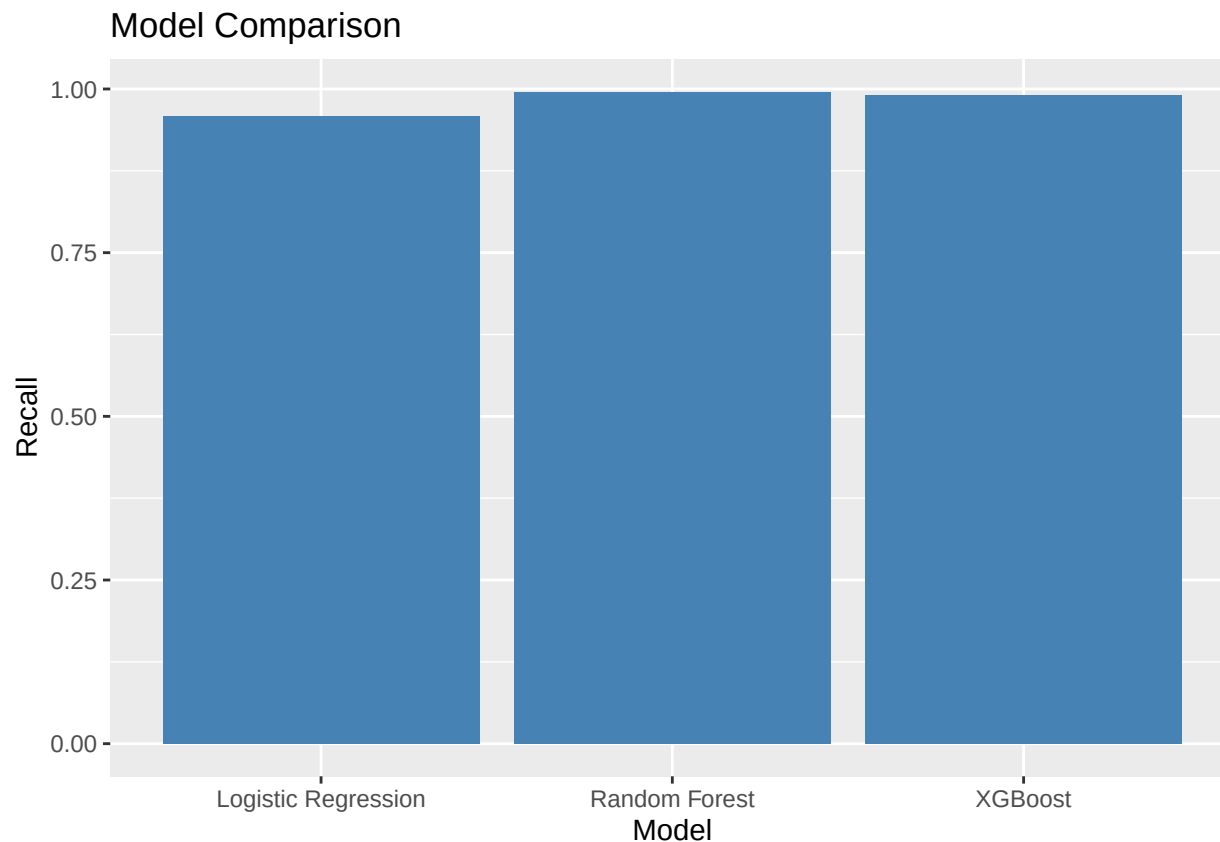
Table 1: Model Comparison Table

Model	Accuracy	Precision	Recall
Logistic Regression	0.790625	0.8013937	0.9583333
Random Forest	0.980000	0.9779051	0.9958333
XGBoost	0.974375	0.9753897	0.9908333

The comparison reveals that Logistic Regression does not perform well because it has linear assumptions

resulting in increased misclassification. Conversely, Random Forest has better performance because it models complex patterns of EEG. The generalization capability is high as indicated by high accuracy and balanced values of precision-recall.

2.10 Result Visualization



2.11 Evaluation Metric

The performance of the classification models is evaluated using Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

3. Results

The results show that Random Forest achieved the highest accuracy (98.18%), followed by XGBoost (97.43%). Logistic Regression did worse (79.12%), which means that the nonlinear models are more adequate to EEG-based seizure detection.

4. Conclusion

This study demonstrates that machine learning models can effectively detect epileptic seizures using EEG data. The highest accuracy of the evaluated models was shown to be random forest (~98%), which means that this model could be used in clinical decision support systems.

Nonetheless, the main weakness is the inter-patient variability of EEG signals, which can impact model generalization in real-world implementation. Moreover, the paper is based solely on time-domain features.

Future research should aim to include frequency-domain analyses like Fourier Transform and Wavelet analysis in order to capture deeper neurological patterns and to calibrate person-specific models to enhance clinical reliability.

5. References

UCI Machine Learning Repository (2025). BEED: Bangalore EEG Epilepsy Dataset. Available at: <https://archive.ics.uci.edu/dataset/1134/beed:+bangalore+eeg+epilepsy+dataset>.